SUMMARY

Chemistry plays a major role in providing needs of human life in our day-to-day life. All things that we come across in life are made of matter. Anything that has mass and occupies space is called matter. Matter is classified based on the physical state and by chemical composition. An element consists of only one type of atom. Compounds contain two or more atoms of same or different elements and their properties are different from those of its constituent elements.

Atoms are too small to measure their masses directly. The IUPAC introduced relative scale of mass based on a standard atom C-12. One twelfth of the mass of a Carbon-12 atom in its ground state is called as Unified atomic mass. 1 amu (or) $1u \approx 1.6605 \times 10^{-27}$ kg. Relative atomic mass is defined as the ratio of the average atomic mass to the unified atomic mass unit. Average atomic mass of an element is the average of the atomic masses of all its naturally occurring isotopes. Molecular mass is the ratio of the mass of a molecule to the unified atomic mass unit. Relative molecular mass is obtained by adding the relative atomic masses of its constituent atoms.

Amounts of substances are usually expressed in moles. A mole is the amount of substance which contains as many elementary entities as there are in 12 gram of Carbon- 12 isotope. Avogadro number is the total number of entities present in one mole of any substance and is equal to 6.022×10^{23}. Molar mass is the mass of one mole of that substance expressed in $g\ mol^{-1}$. One mole of an ideal gas occupies a volume of 22.4 litre at 273 K and 1 atm pressure. Similar to the mole concept, the concept of equivalent mass is also used in analytical chemistry. Gram equivalent mass is the mass of an element (compound/ion) that combines or displaces 1.008 g hydrogen, 8 g oxygen or 35.5 g chlorine. Elemental analysis of a compound gives the mass percentage of atoms from which empirical and molecular formula are calculated. Empirical formula is the simplest ratio of the number of different atoms present in one molecule of the compound. Molecular formula is the formula written with the actual number of different atoms present in one molecule. A quantitative relationship between reactants and products can be understood from stoichiometry. Stoichiometry gives the numerical relationship between chemical quantities in a balanced equation. When a reaction is carried out using non-stoichiometric quantities of the reactants, the product yield will be determined by the reactant that is completely consumed and is called the limiting reagent. It limits the further reaction to take place. The other reagent which is in excess is called the excess reagent.

The reaction involving loss of electron is oxidation and gain of electron is reduction. Usually both these reactions take place simultaneously and are called as redox reactions. These redox reactions can be explained using oxidation number concept. Oxidation number is the imaginary charge left on the atom when all other atoms of the compound have

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